

Development of net energy intake models for drift-feeding juvenile coho salmon and steelhead

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Received: 6 June 2007 / Accepted: 14 November 2007 / Published online: 26 January 2008
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Abstract We developed models to predict the effect of water velocity on prey capture rates and on optimal foraging velocities of two sympatric juvenile salmonids, coho salmon and steelhead. Mean fish size was ~80 mm, the size of age I+ coho and steelhead during their second summer in Southeast Alaska streams, when size overlap suggests that competition might be strongest. We used experimentally determined prey capture probabilities to estimate the effect of water

velocity on gross energy intake rates, and we modeled prey capture costs using experimental data for search and handling times and published models of swimming costs. We used the difference between gross energy intake and prey capture costs to predict velocities at which each species maximized net energy intake rate. Predicted prey capture rates for both species declined from ~75 to 30–40 prey/h with a velocity increase from 0.30 to 0.60 m·s⁻¹. We found little difference between coho and steelhead in predicted optimum foraging velocities (0.29 m·s⁻¹ for coho and 0.30 m·s⁻¹ for steelhead). Although prey capture ability appears to be more important than are prey capture costs in determining optimum foraging velocities, capture costs may be important for models that predict fish growth. Because coho are assumed to pay a greater swimming cost due to a less hydrodynamic body form, we also modeled 10 and 25% increases in hydrodynamic drag to assess the effect of increased prey capture costs. This reduced optimum velocity by 0 and 0.01 m·s⁻¹, respectively. Habitat segregation among equal-sized coho and steelhead does not appear to be related to the effects of water velocity on their respective foraging abilities.

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Keywords Drift-feeding · Habitat selection ·
Net energy intake · *Oncorhynchus kisutch* · *O. mykiss*
irideus · Water velocity

Introduction

Cost-benefit analyses have played a central role in the development of foraging- and habitat-selection theory in ecology (Holling 1959; MacArthur and Pianka 1966; Charnov 1976; Stephens and Krebs 1986). For drift-feeding salmonids, the metabolic cost of foraging against a current is balanced against the benefit of more prey encounters in faster water (Everest and Chapman 1972; Wankowski 1981; Fausch 1984). Cost-benefit curves have been proposed for drift-feeding salmonids, wherein the fish's gross energy intake (GEI) is determined by the number and type of prey it captures, and its net energy intake (NEI) is determined by its GEI minus the energetic costs (EC) of prey capture (Hughes and Dill 1990; Hill and Grossman 1993; Nislow et al. 1999). Based on this theory, NEI models have been developed to predict energy intake, growth, and distribution for stream salmonids (Hayes et al. 2000; Guensch et al. 2001; Hughes et al. 2003). These models have also begun to be used to assess how stream habitat and management influences salmonid populations (Nislow et al. 1999). The effects of habitat characteristics such as water velocity on foraging costs and prey capture success, however, are still poorly understood (Hughes et al. 2003; Piccolo et al. 2007).

Water velocity is a key component of drift-foraging models because: (1) it determines the rate at which fish encounter prey (i.e., their encounter rate), (2) it influences their ability to capture prey (i.e. their capture probability), and (3) it influences the costs, in terms of both energy and time, that the fish incurs while capturing prey. In order to make accurate predictions therefore, drift-foraging models need to incorporate realistic relationships between water velocity and encounter rate, capture probability, and capture costs (Hughes et al. 2003). Prey encounter rate is usually assumed to increase linearly with water velocity (Nislow et al. 1999; Hayes et al. 2000); capture probability has either been assumed to be 100% (Hughes and Dill 1990), or it has been modeled based on pilot experiments (Hill and Grossman 1993; Nislow et al. 1999); capture costs have typically been approximated as steady swimming costs, ignoring the potential increased costs of pursuing and capturing prey (Hill and Grossman 1993; Nislow et al. 1999; Hughes et al. 2003). In Piccolo et al. (2008) and in this paper, we attempt to improve our mechanistic

understanding of the effects of water velocity on capture probability and prey capture costs by using detailed 3-D analysis to better define the costs and benefits of prey capture.

Prey capture probability for drift-feeding salmonids has been shown to decrease with increasing water velocity (Hill and Grossman 1993; Nislow et al. 1999; Piccolo et al. 2008). Theory predicts that a fish's optimum foraging velocity will be that at which it maximizes its NEI, i.e., the velocity at which decreased prey capture probability outweighs the benefit of more prey being delivered by faster current (Hill and Grossman 1993; Nislow et al. 1999). It follows that two co-evolved sympatric species might reduce niche overlap by maximizing NEI at different water velocities. In this paper we present the first experimentally-derived cost-benefit curves for two sympatric drift-feeding stream salmonids, juvenile coho salmon (*Oncorhynchus kisutch* (Walbaum)) and steelhead [*Oncorhynchus mykiss irideus* (Gibbons)]. Juvenile coho and steelhead have been shown to segregate stream habitat, with coho using slower-velocity, deeper pools and steelhead using faster-velocity, shallower riffles (Hartman 1965; Allee 1981; Bugert et al. 1991). Bisson et al. (1988) suggested that this habitat segregation might be due to morphology-dependent differences in foraging abilities. Coho are more laterally compressed with taller fins, and steelhead are more cylindrical with shorter fins. This would allow coho to forage more efficiently in slower water, where fast turning and acceleration abilities are favored because prey are more patchily distributed. The steelhead body form would be better adapted to minimize drag while drift-feeding in flowing water. Differences in morphology have been shown to influence feeding performance, competitive ability, and habitat selection in other sympatric pairs of fish species including sunfish (Werner 1977; Huckins 1997) and sticklebacks (Lavin and McPhail 1986). To date, this hypothesis has not been tested for two sympatric species of stream salmonids

Our objective was to develop NEI models that predict optimum foraging velocities for coho and steelhead. We used the results of laboratory experiments to model gross energy intake and prey capture costs, and we estimated NEI as the difference between these curves. We used NEI to predict optimum foraging velocities, to test the hypothesis that coho have a slower optimum velocity than do steelhead.

We also assessed how 10 and 25% increases in hydrodynamic drag, and associated foraging costs for coho (as suggested by body form) influence NEI rate and optimal foraging velocity.

Materials and methods

Foraging experiments were conducted in a variable-velocity stream tank (Piccolo et al. 2008), and were videotaped with two cameras to allow for 3-D analysis (Hughes and Kelly 1996a) (Fig. 1). We used five wild fish of each species, 75–80 mm fork length. Mean water temperature was 10.46°C (SD=0.84); mean dissolved oxygen was 7.78 mg·l⁻¹ (SD=0.66); mean pH was 7.9 (SD=0.21); mean turbidity was 0.36 NTU (SD=0.16). We selected five water velocity treatments (0.29, 0.39, 0.48, 0.54, and 0.61 m·s⁻¹ mean water column velocity), ranging from the slowest at which fish would continually hold station and drift feed, to near the maximum published value for 75-mm juvenile coho and steelhead (Everest and Chapman 1972; Beecher et al. 1993). Each fish was fed 100 individual prey (2-mm frozen brine shrimp) over a 25 min period. The number of prey captured was determined from the video tapes. Prey capture maneuvers were digitized to obtain 3-D coordinates (Hughes and Kelly 1996a), providing measurements of the effects of water velocity on search and handling times, prey interception distances, and swimming speeds. Fig. 1 provides an example of the results of the 3-D data analysis. See Piccolo et al. (2008) for a

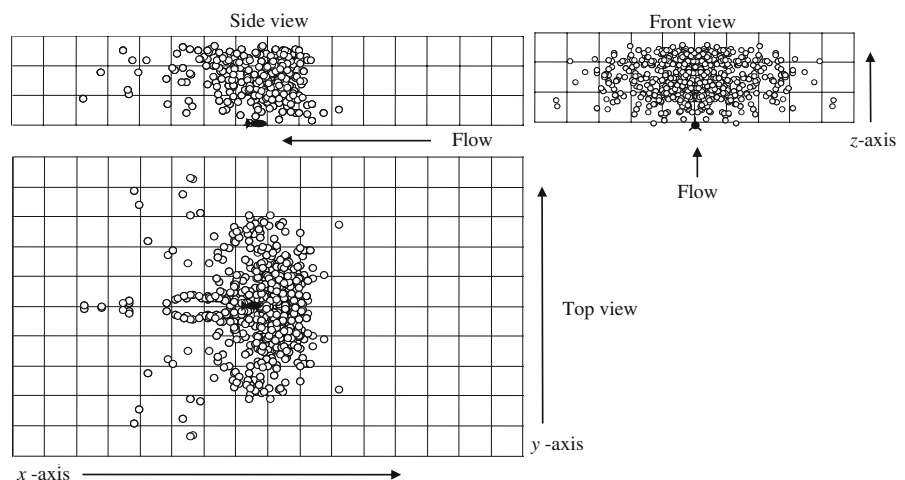
detailed description of the stream tank, experimental protocol, and 3-D data analysis.

We used linear regressions of the number of prey captures *vs* water velocity (Fig. 2a) to predict prey capture rates for coho and steelhead (Fig. 2b). For each regression there was a significant negative relationship between water velocity and the number of prey captures ($P<0.001$), with no significant difference between species (Piccolo et al. 2008). For the foraging experiments we held prey encounter rate constant (4 prey·min⁻¹) across all velocity treatments to minimize the possibility of an interaction between water velocity and prey handling time (Piccolo et al. 2008). This resulted in a decrease in prey density (number of prey · m⁻³) as velocity increased, because the volume of water increased while the number of prey was constant. In streams, however, prey encounter rates are expected to increase proportionally with water velocity (Everest and Chapman 1972; Wankowski and Thorpe 1979). To correct for prey density, we predict prey capture rate as follows:

$$\text{Prey capture rate} = [\text{Number of prey captured}] \times [\text{Treatment velocity}/\text{Fastest velocity}]$$

This gives the density-corrected number of prey captured per 25-min feeding trial, which is then multiplied by 2.4 to predict number of prey per hour. We plotted density-corrected prey capture rate *vs* water velocity for all fish from the foraging experiments (Fig. 2b). We then plotted density-corrected capture curves for each species by predicting capture

Fig. 1 Front, side, and top view of prey capture locations for coho at 0.29 m·s⁻¹ water velocity. Each open circle represents an individual prey capture, and the dark fish figure represents the focal point. Data is pooled for five fish, each of which was tested individually, as shown by the regression in Fig. 2a. Estimates of energetic costs (EC) are derived from this 3-D analysis



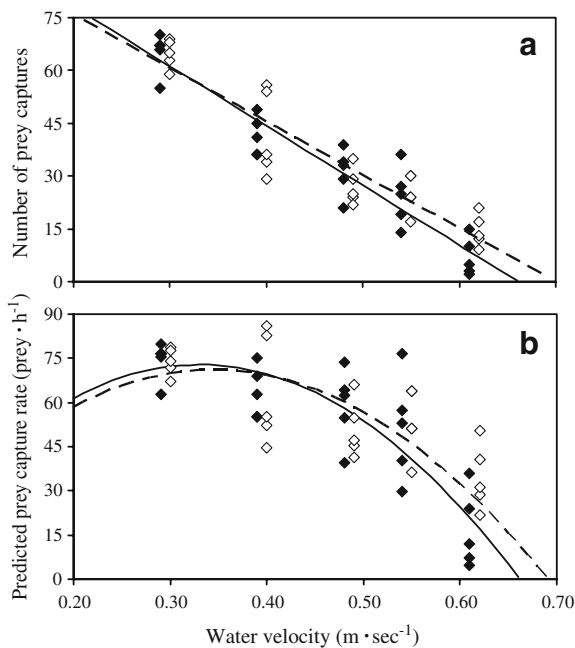


Fig. 2 **a** The number of prey captures per 25-min foraging experiment vs water velocity for coho (solid diamonds, line) and steelhead (open diamonds, dashed line). **b** Predicted prey capture rate vs water velocity for coho (solid diamonds and line) and steelhead (open diamonds, dashed line). Diamonds are the predicted prey capture rates for individual coho and steelhead ($n=5$ for each species), and lines are predicted from the linear regressions from (a). In both panels steelhead data points are offset $+0.01 \text{ m} \cdot \text{s}^{-1}$ for visual clarity

rate at 0.01-m intervals using the regression equations from Piccolo et al. (2008):

$$\text{For coho : } y = -169.2x + 112.0$$

$$\text{For steelhead : } y = -151.9x + 104.8$$

Where y =the number of prey captured and x =water velocity ($\text{m} \cdot \text{s}^{-1}$).

To estimate NEI ($\text{J} \cdot \text{h}^{-1}$) we used the following model (Table 1):

$$\text{NEI} = \text{GEI} - \text{EC} \quad (1)$$

Where GEI is gross energy intake rate ($\text{J} \cdot \text{h}^{-1}$), and EC is energetic cost ($\text{J} \cdot \text{h}^{-1}$). We predicted GEI vs water velocity for each species by substituting the mean energy content of a 2-mm mayfly (Cummins and Wuycheck 1971) for that of our 2-mm brine shrimp. The cut brine shrimp were not dissimilar in size or shape to a 2-mm mayfly. We estimated EC as search time (ST) and costs (SC) plus handling time (HT) and costs (HC). Energetic costs are therefore:

$$\text{EC} = [(\text{ST} \cdot \text{SC}) + (\text{HT} \cdot \text{HC})] \quad (2)$$

Where ST is the time spent at the focal point and HT is the time spent pursuing prey and returning to the focal

Table 1 Parameters used in the net energy intake models for coho and steelhead

Parameter	Units	Equation
Net energy intake (NEI)	$\text{J} \cdot \text{h}^{-1}$	$\text{NEI} = \text{GEI} - \text{EC}$
Gross energy intake (GEI)	$\text{J} \cdot \text{h}^{-1}$	$\text{GEI} = \text{energetic content of prey (J)} \cdot \text{number of prey captured} \cdot \text{h}^{-1}$
Energetic cost (EC)	$\text{J} \cdot \text{h}^{-1}$	$\text{EC} = [(\text{ST} \cdot \text{SC}) + (\text{HT} \cdot \text{HC})]$
Search time (ST)	h	$[\text{Total time (TT) of a feeding trial (25 m)} - (\text{IT} + \text{RT})] \cdot 60$
Interception time (IT)	h	The median time taken for the fish to swim from the focal point to the point of prey capture; calculated independently for each species at each velocity treatment
Return time (RT)	h	The median time taken for the fish to swim from the point of prey capture back to the focal point; calculated independently for each species at each velocity treatment
Search costs (SC)	$\text{J} \cdot \text{h}^{-1}$	The energetic cost of maintaining a focal point position by steady swimming $[(\text{mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}) \cdot \text{ST}]$; derived from Brett and Glass (1973).
Handling time (HT)	h	$\text{IT} + \text{RT}$
Handling costs (HC)	$\text{J} \cdot \text{h}^{-1}$	The energetic cost of swimming to capture a prey $[(\text{mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}) \cdot \text{HT}]$. Calculated as steady swimming costs (Brett and Glass 1973) multiplied by drag factor to simulate unsteady swimming (Hughes and Kelly 1996b)

Parameters were measured experimentally by Piccolo et al. (2008) except where noted

point. We designated the median search time for each species at each velocity as the total time (TT) of the experiment minus the sum of the median interception time (IT) plus the median return time (RT):

$$ST = TT - (IT + RT) \quad (3)$$

We estimated the cost of ST as a *steady* swimming cost, because when a fish is holding station at the focal point it swims in a relatively straight line and experiences equal hydrodynamic thrust and drag (Hughes and Kelly 1996b). To estimate metabolic rates ($\text{mg O}_2 \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$) we used the temperature-, size-, and swimming speed-specific relationships developed by Brett and Glass (1973) for juvenile sockeye salmon. We used this rate for both species because it is the only relationship developed for juvenile salmon that covers the fish sizes, temperatures, and swimming speeds we used. Puckett and Dill (1985) reported that metabolic rates for sustained swimming in coho (40 mm, 15°C) closely matched those of sockeye under similar conditions, and that the sockeye relationship might be acceptable for coho. Morgan and Iwama (1991) reported that metabolic rates for juvenile steelhead were comparable to those reported by Brett and Glass (1973) for sockeye of similar size and activity.

Because there is strong evidence that swimming costs are significantly greater during prey capture maneuvers than they are during search time at the focal point (Boisclair and Tang 1993), we estimated the costs of prey interception and return as *unsteady* swimming costs, based on a model developed by Hughes and Kelly (1996b). The model accounts for the additional hydrodynamic drag that a fish encounters during the swimming associated with prey capture maneuvers. During the duration of each prey capture maneuver (i.e. HT), therefore, drag is increased by a factor of 3, resulting in greater swimming costs relative to those during time spent at the focal point [see Hughes and Kelly (1996b) for a detailed explanation].

To assess the effects of increased swimming costs on optimal velocity for coho (based on body morphology, i.e. Bisson et al. (1988), we used 10 and 25% increases in the drag multiplier for both steady and unsteady swimming costs (i.e. ST and HT).

We predicted the optimum foraging velocity for each species, and for coho under increased swimming

costs, as the velocity at which the difference between GEI and EC (i.e. NEI) was greatest.

Results

Predicted prey capture rates peaked at 73 and 71 $\text{prey} \cdot \text{h}^{-1}$ for coho and steelhead, respectively (Fig. 2b). Fish could attain 95% of their peak prey capture rates at velocity ranges from 0.25–0.40 and 0.26–0.42 $\text{m} \cdot \text{s}^{-1}$ for coho and steelhead, respectively. Predicted gross energy intake rate peaked at 70.0 $\text{J} \cdot \text{h}^{-1}$ (0.33 $\text{m} \cdot \text{s}^{-1}$) for coho, and at 68.4 $\text{J} \cdot \text{h}^{-1}$ (34 $\text{m} \cdot \text{s}^{-1}$) for steelhead (Fig. 3).

Predicted optimum foraging velocity was 0.29 $\text{m} \cdot \text{s}^{-1}$ ($\text{NEI} = 51.2 \text{ J} \cdot \text{h}^{-1}$) for coho and 0.30 $\text{m} \cdot \text{s}^{-1}$ ($\text{NEI} = 50.4 \text{ J} \cdot \text{h}^{-1}$) for steelhead (Fig. 3). The ranges where each species could obtain 95% of NEI were 0.24–0.34 and 0.25–0.36 $\text{m} \cdot \text{s}^{-1}$ for coho and steelhead respectively.

Predicted prey capture costs were similar for coho and steelhead (Fig. 3) ranging from $\sim 6 \text{ J} \cdot \text{h}^{-1}$ at slow velocities (i.e. basal metabolic rate) to $\sim 40 \text{ J} \cdot \text{h}^{-1}$. At optimum foraging velocities costs were 17.9 and 16.8 $\text{J} \cdot \text{h}^{-1}$ for coho and steelhead, respectively, which equals $\sim 25\%$ of GEI for each species. Predicted costs rose exponentially vs water velocity until they reached the velocity at which they crossed the GEI line. After this they declined sharply because fish were pursuing few prey, so costs were essentially only steady swimming costs at the focal point.

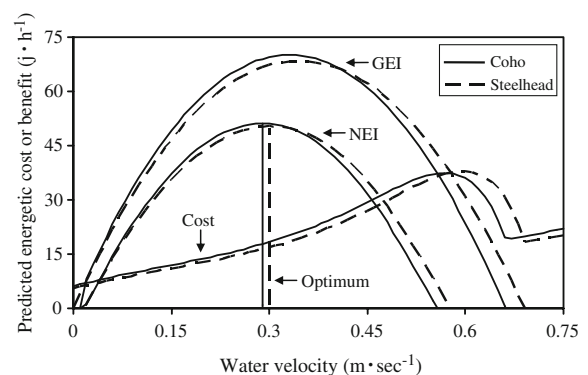


Fig. 3 Estimated gross energy intake (GEI), net energy intake (NEI), and costs vs water velocity for coho (solid lines) and steelhead (dashed lines). Vertical lines indicate optimum foraging velocity for coho (solid) and steelhead (dashed). The steep declines in the cost curves at fast velocities occur because, as capture rates fall to zero, costs are only those of holding station at the focal point

We used the same equations to predict basal metabolic rates for both species. For their respective cost curves to differ, therefore, coho and steelhead would have needed to differ in the proportion of time spent at the focal point (i.e. relatively low-cost, steady swimming) vs that of intercepting prey (higher-cost, unsteady swimming). These proportions were, in fact, very similar between coho and steelhead (Piccolo 2008).

Increasing the drag multiplier for coho by 10 and 25% during both steady and unsteady swimming (i.e. ST and HT) resulted in decreases in optimum foraging velocity of 0.0 and 0.01 $\text{m}\cdot\text{s}^{-1}$, respectively. Net energy intake was reduced from 50.4 $\text{J}\cdot\text{h}^{-1}$ without an increase of the drag multiplier, to 49.8 and 47.5 $\text{J}\cdot\text{h}^{-1}$ at 10 and 25% increases, respectively (Fig. 4). Prey capture costs were 17.9, 19.2, and 20.9 $\text{J}\cdot\text{h}^{-1}$ at 0, 10, and 25% increases in drag multiplier, respectively (Fig. 4).

Discussion

Optimal foraging velocities for coho (0.29 $\text{m}\cdot\text{s}^{-1}$) and steelhead (0.30 $\text{m}\cdot\text{s}^{-1}$) were found to be nearly equal to that predicted by Hill and Grossman's (1993) model for similar-sized rainbow trout (0.28 $\text{m}\cdot\text{s}^{-1}$) under similar conditions. Our predicted NEI rates were lower than those of Hill and Grossman (1993),

because we used smaller-sized prey, and possibly because of lower prey densities (i.e. lower encounter rates). Hill and Grossman (1993) do not report prey density or the size distribution of prey for their model. Our models also agree with previous models that predict prey capture costs are relatively low in relation to GEI (Hughes and Dill 1990; Hill and Grossman 1993). Capture costs at optimal velocity in our model do represent a greater proportion of GEI (~25%) than do those in Hill and Grossman (1993) (~13%). This is likely due to the fact that we included the costs of unsteady swimming during prey capture maneuvers. Because our estimates are based on accurate assessment of the times and distances of capture maneuvers, they may be more realistic approximations of drift foraging costs. Regardless, capture costs appear to have less influence on optimal foraging velocity than does prey capture ability (Hughes and Dill 1990; Hill and Grossman 1993; Piccolo et al. 2008; this paper).

It is interesting that our model predicts a similar optimal foraging velocity to that of Hill and Grossman (1993), despite differences in prey sizes, and differences in how prey capture probability was estimated. Hill and Grossman (1993) included only prey ≥ 2 mm, whereas our model uses only 2-mm prey. Because fish detect larger prey further away, including larger should increase the foraging area, with an attendant increase in prey encounters. If fish are unable to capture more of these prey because of limitations in capture ability, however, increasing prey encounter rate would raise the GEI curve higher, but not shift it to the right (i.e. increase optimal velocity). Hill and Grossman's (1993) prediction of higher GEI, but similar optimal velocities, seems to support the idea that capture probability, not capture costs, limit optimum velocity. Piccolo et al. (2008) suggest that the mechanism limiting capture probability appears to be prey-detection limitations. Further model development would benefit from experiments controlling for different prey sizes.

By including the unsteady swimming costs of prey capture in our model we estimated the energetic costs at optimal velocity to be approximately double that of Hill and Grossman's (1993) model, in which capture costs were ignored. Although the increased costs did not shift the NEI curve to the left, they did lower NEI at the optimum velocity, which would reduce the energy available for growth or reproduction. This may have important consequences for models that predict

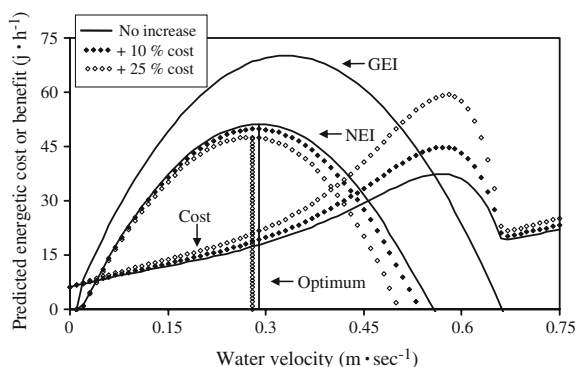


Fig. 4 Estimated gross energy intake (GEI), net energy intake (NEI) and costs vs water velocity for coho under three different cost scenarios relative to steelhead costs: (1) equal to steelhead (no increase, solid lines), (2) 10% increase in drag factor, solid diamonds, and (3) 25% increase in drag factor, open diamonds. Vertical lines indicate optimum foraging velocities (no increase and +10% are superimposed). The steep declines in the cost curves at fast velocities occur because, as capture rates fall to zero, costs are only those of holding station at the focal point

fish growth (e.g. Hayes et al. 2000); it may be less important for models used to predict fish distribution (e.g. Hughes and Dill 1990). Hayes et al. (2000) model included an estimate of prey capture costs, and they found that it accurately predicted whole-lifetime growth patterns for brown trout. Other models have successfully predicted habitat selection without including prey capture costs (Hughes and Dill 1990; Nislow et al. 1999; Hughes et al. 2003). Our 3-D analysis of foraging maneuvers should represent an accurate assessment of the energetic costs of prey capture, and hence they have the potential to improve estimates of NEI.

Modeled increases in foraging costs for coho resulted in virtually no change in optimum velocity, and little change in NEI at optimum velocity. This, combined with similar relationships between prey capture success and water velocity for coho and steelhead (Piccolo et al. 2008), suggest that differential foraging abilities (i.e. Bisson et al. 1988) may not be responsible for velocity segregation between the species. We have also found that depth-dependent differences in foraging abilities are unlikely to explain habitat segregation (Piccolo et al. 2007). We hypothesize, therefore, that segregation is based on either (1) other selective mechanisms, such as prey selectivity or life history attributes (e.g. different growth trajectories and therefore different energetic demands), or (2) interactive mechanisms such as competitive exclusion.

Most of the research on coho and steelhead habitat segregation has been conducted in the Pacific Northwest and British Columbia, where there is little size overlap between age 0+ coho and steelhead due to differences in emergence times (Hartman 1965; Fraser 1969; Allee 1981; Young 2004). Hartman (1965) and Fraser (1969) found that steelhead grow faster than do coho, and they are equal in size by the end of the summer. In these systems, coho usually smolt the following spring, thus avoiding potential competition when sizes overlap. In Southeast Alaska, coho remain in streams for a second growing season and smolt at age II+, and steelhead remain for at least a third summer and smolt at age III+ (Lohr and Bryant 1999; Halupka et al. 2000). Because steelhead grow faster than do coho, fish of both species are the same size (~75 mm) at the start of their second summer at age I. We have observed instances in two separate streams in which age I+ steelhead were actively foraging during the day, while age I+ coho were congregated

in deep pools and foraging very little (Piccolo, unpublished data). Age 0+ coho were foraging in close proximity to the age I+ steelhead in both instances, which suggests that the age I+ coho may have chosen not to forage. It is possible that growth requirements of age I+ coho and steelhead differ, and that age I+ coho accept a slower growth rate in exchange for less predation risk because they will smolt early the following summer. This has been suggested for coho (Reinhardt 1999) and clearly demonstrated for Atlantic salmon (Thorpe et al. 1992; Metcalfe 1998; Juanes et al. 2000). If this is the case it would be an example of selective segregation based on long-term survival, rather than on short-term energetic considerations. The temporal and spatial persistence of this pattern for coho and steelhead is undocumented.

Although equal-sized coho and steelhead appear to have remarkably similar foraging abilities, we have found slight advantages for coho in deeper water (Piccolo et al. 2007) and for steelhead in faster water (Piccolo et al. 2008). These differences occurred only at the deepest and fastest treatments, respectively, which are probably at or near the maximum depths and velocities at which fish of this size forage. Nonetheless, they might be of importance during periods of resource limitation. We used only one type and size of prey in our experiments, and differential foraging abilities related to these factors (e.g. aquatic vs terrestrial prey) might also facilitate coexistence. Nakano et al. (1999) demonstrated that coexistence of two species of Japanese charr was facilitated by a foraging-mode shift by one species from drifting to benthic prey during periods of prey limitations. Specialization by prey type and location in the water column has been suggested for coho and steelhead (Johnson and Ringler 1980), but there have been no conclusive experiments. More intensive field studies and experiments are needed to clearly document habitat selection by coho and steelhead. These studies will need to account for fish size, and include measures of habitat and prey availability as well as direct observations of habitat use, foraging, and agonistic behavior.

Acknowledgements Funding and support was for this project was provided by: The University of Alaska Rasmuson Fisheries Research Center and the USDA Pacific Northwest Research Station. Two anonymous reviewers greatly improved an earlier version of this manuscript.

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